

Big Data, Data Roles, Data Cleaning, Normalization, and Hypothesis Testing

1. Data Roles

Data Analyst analyzes reports and dashboards. Data Scientist builds prediction models. ML Engineer deploys and optimizes models. Data Engineer builds pipelines and data systems.

2. Big Data Characteristics (5Vs)

Volume: huge amount of data. Velocity: speed of data generation. Variety: different types of data. Veracity: data quality and trust. Value: useful business insights.

3. Messy Data vs Noisy Data

Messy data is unorganized or inconsistent such as 'Egypt', 'egypt', 'EGYPT'. Noisy data contains wrong values such as age = 999.

4. Binning

Sort data, divide into bins, then replace values using mean, median, or boundaries.

5. Handling Spelling Errors

Use lowercase conversion, remove spaces, replace incorrect values, and fuzzy matching.

6. Outliers

Detect using IQR or Z-score. Handle by removing, replacing with median, capping, or transformation.

7. Data Integration Methods

Entity identification, redundancy detection, value conflict detection, and correlation analysis.

8. Max-Min Normalization

$$X' = (X - X_{\min}) / (X_{\max} - X_{\min})$$

9. Z-score Normalization

$$Z = (X - \text{mean}) / \text{standard deviation}$$

10. t-distribution

Used when sample size is small and population standard deviation is unknown.

11. f-distribution

Used in ANOVA and comparing multiple group means.

12. Hypothesis Testing Fields

Used in medicine, marketing, education, industry, machine learning, economics, and banking.

13. Z-test and T-test

Z-test is used for large samples. T-test is used for small samples.

14. Decision Rule

If $p\text{-value} < 0.05$ reject Null Hypothesis. Otherwise fail to reject Null Hypothesis.